

Advances in Binocular Stereo Vision-Driven 3D Perception and Intelligent Analysis Methods for Agriculture

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Abstract

Binocular stereo vision is a low-cost and scalable 3D perception technology that shows strong potential in agricultural phenotyping and smart agriculture. By estimating depth from multi-view RGB images, it enables non-contact, high-precision sensing of crop structure, canopy morphology, growth dynamics, and livestock traits, providing essential support for digital and intelligent agricultural production. With recent advances in deep learning-based stereo matching, multimodal sensor fusion, and 3D reconstruction, its robustness and accuracy in complex field environments have been significantly improved. This paper systematically reviews recent progress in agricultural applications of binocular stereo vision, covering system architectures, traditional and deep learning-based stereo matching methods, point cloud reconstruction techniques, and emerging supervision strategies such as 3D Gaussian splatting. It further summarizes key applications, including high-throughput phenotyping, fruit localization and robotic harvesting, weed detection and precision spraying, autonomous navigation, and livestock body condition assessment, highlighting its role in multi-task agricultural perception systems. Finally, the paper discusses major challenges, including low-texture matching difficulty, occlusions in complex environments, cross-domain generalization, real-time lightweight deployment, and limited dataset availability. Future directions are outlined in foundation model-based visual perception, self- and weakly supervised learning, multimodal fusion, and edge-efficient model design, aiming to support large-scale deployment in smart agriculture.

Keywords: binocular stereo vision; agricultural 3D perception; deep learning; smart agriculture; multimodal fusion

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1. Introduction

Plant phenotypes refer to the set of observable traits resulting from the interaction between genotype and environment. Accurate acquisition of crop phenotypic information is fundamental for understanding genotype–phenotype–environment interactions and serves as an essential basis for modern crop breeding, precision crop management, and intelligent agricultural decision-making [1–3]. Traditional phenotyping methods rely heavily on manual measurements, such as plant height estimation with rulers, leaf counting, and destructive biomass sampling [4,5]. As shown in the study by Zou et al., manual

counting in mature wheat density estimation is not only inefficient, but also incurs substantially increasing labor costs as cultivation scale expands [6]. In addition, conventional approaches typically depend on discrete sampling at isolated time points, making it difficult to capture continuous growth dynamics throughout the crop life cycle, which often leads to missing information at key developmental stages.

Two-dimensional (2D) imaging technologies have partially alleviated the limitations of manual measurements. Unmanned aerial vehicles (UAVs) and satellite remote sensing platforms enable large-scale and high-throughput data acquisition [7,8]. However, due to the inherent loss of depth information in 2D projections, the full three-dimensional structure of crops cannot be faithfully represented. Issues such as leaf overlap, organ occlusion, and viewpoint distortion remain difficult to resolve [9–11]. For crops with complex canopy structures, such as maize, wheat, and rice, 2D images are insufficient to accurately characterize key structural traits, including leaf area index, leaf angle distribution, and canopy volume, thereby limiting their applicability in fine-grained phenotypic analysis [12,13].

In parallel with advances in sensing hardware, intelligent analysis methods have become increasingly important for converting agricultural images and three-dimensional data into biologically and operationally meaningful information. Conventional machine-learning methods rely on hand-crafted features and statistical classifiers, whereas recent deep learning approaches can automatically learn hierarchical representations for image classification, object detection, semantic and instance segmentation, stereo matching, and pose estimation. For three-dimensional agricultural data, point-cloud filtering, registration, segmentation, skeleton extraction, geometric modeling, and trait-estimation algorithms are commonly used to characterize plant and animal structures. Multimodal sensor fusion and temporal modeling further integrate complementary information from RGB images, depth sensors, LiDAR, multispectral cameras, and sequential observations. Together, these intelligent analysis methods support crop phenotyping, fruit localization, weed recognition, autonomous navigation, robotic harvesting, and livestock monitoring, thereby transforming raw sensor measurements into actionable information for smart agriculture.

From a global perspective, binocular stereo vision and intelligent analysis methods have attracted increasing attention in agricultural research and automation. Studies in North America, Europe, and East Asia have progressively moved from controlled laboratory experiments toward field-based deployment and integrated perception systems for high-throughput phenotyping, robotic harvesting, precision spraying, autonomous machinery, and livestock monitoring. Regional research priorities vary according to crop types, production systems, labor availability, equipment infrastructure, and investment in agricultural digitalization. Meanwhile, the availability of low-cost stereo cameras, deep learning models, edge-computing platforms, and multimodal sensing systems has lowered the technical barriers to agricultural 3D perception. Nevertheless, global adoption remains uneven because of limited domain-specific datasets, substantial environmental variability, differences in farm scale and infrastructure, and the absence of standardized evaluation and deployment frameworks. These conditions indicate that future progress requires not only algorithmic innovation, but also internationally diverse datasets, cross-region validation, affordable hardware integration, and closer collaboration among researchers, equipment manufacturers, and agricultural practitioners.

The emergence of three-dimensional (3D) perception technologies has provided a promising pathway to overcome these limitations. Point clouds and mesh-based representations preserve complete spatial geometric information, enabling organ-level morphometric measurements and canopy-scale structural analysis [14,15]. Wu et al. con-

structured maize plant point clouds using multi-view stereo vision and achieved the automatic extraction of stem–leaf segmentation, plant height, and leaf area, with a coefficient of determination (R^2) of 0.994 between estimated and manually measured plant height [16]. Zhou et al. further developed a temporal point cloud dataset covering four crops, including maize, soybean, and rice, across eleven varieties [17]. By combining skeleton extraction and leaf matching, they enabled cross-temporal organ tracking and revealed dynamic growth patterns across different crops. Compared with 2D methods, 3D data provide richer structural information and support non-destructive, continuous monitoring, significantly improving measurement consistency and repeatability [18,19].

Among existing 3D sensing technologies, binocular stereo vision occupies a unique position. Current crop 3D sensing approaches can be broadly categorized into active and passive methods [20–22]. Active methods, including structured light, time-of-flight (Time of Flight) cameras, and LiDAR, obtain depth information by actively emitting signals [23,24]. Although these methods offer high accuracy, they suffer from high hardware cost and limited adaptability to outdoor environments. In field-based crop phenotyping, ToF cameras are prone to generating large invalid measurement regions under strong ambient illumination, while LiDAR, despite its high precision, remains expensive. Passive reconstruction methods mainly include multi-view stereo (Multi-View Stereo), structure from motion (Structure from Motion), and binocular stereo vision, all of which rely solely on ambient light for 3D reconstruction [25–27].

Binocular stereo vision has become one of the most promising solutions for agricultural 3D perception due to several advantages [28,29]. First, it is highly cost-effective, requiring only two standard RGB cameras, with hardware costs far lower than LiDAR systems [30]. Zhang et al. compared fruit tree canopy volume estimation between binocular vision and LiDAR in UAV-based precision spraying and reported that the binocular system reduced cost by more than 80% while maintaining volume estimation errors within 10% [31]. Second, binocular stereo vision provides a practical and cost-effective solution for open-field applications, but its performance remains sensitive to strong and spatially variable illumination. Direct sunlight, shadows, overexposure, and specular reflections may reduce photometric consistency between stereo image pairs and consequently degrade matching accuracy. In comparison, LiDAR is generally more robust to variations in ambient illumination, although intense solar radiation may still reduce its signal-to-noise ratio and effective sensing range. Third, it provides high information density by generating dense depth maps, which are essential for fine-grained organ-level phenotyping. Yang et al. combined binocular stereo vision with supervision signals derived from 3D Gaussian splatting to generate dense depth maps for leafy vegetables, achieving an endpoint error (EPE) of only 0.91 pixels in leaf regions [32]. Fourth, the method benefits from a mature algorithmic foundation, and recent advances in deep learning have further significantly improved stereo matching accuracy [33].

However, binocular stereo vision also faces several critical limitations in agricultural applications. The first is the difficulty of stereo matching in low-texture and repetitive-texture regions, as crop leaf surfaces often lack distinctive features while repetitive structures easily lead to mismatches. Wu et al. observed in oil-tea fruit harvesting robots that traditional stereo matching methods produce large holes in leaf regions, whereas improved deep learning-based networks significantly reduce 3D localization errors [16]. Occlusion remains another major challenge. Zou et al. reported that severe occlusion in dense wheat canopies leads to substantial segmentation errors when directly applying clustering methods on reconstructed point clouds [6]. The trade-off between computational complexity and real-time performance is also prominent. Yi et al., in a study on real-time volume estimation of four fruit types, found that lightweight models achieved higher frame rates but underperformed in occlusion handling and small-object detection compared to

more complex networks [34]. In addition, high-quality stereo matching datasets for agricultural scenes are extremely limited. Unlike autonomous driving, which benefits from large-scale datasets such as KITTI, existing agricultural datasets are small and lack diversity. To address this issue, Yang et al. constructed a lettuce stereo dataset with 36 scenes and more than 8000 stereo image pairs by leveraging 3D Gaussian splatting for multi-view reconstruction and dense annotation generation [32].

To address these challenges, recent studies have made substantial progress from the perspectives of algorithm design, dataset construction, and system integration. Niu et al. proposed a stereo foundation model fine-tuning strategy driven by 2D Gaussian splatting for maize field scenarios [35]. Using multi-view reconstruction, they synthesized 3625 high-fidelity stereo pairs, and through a partial parameter freezing strategy, fine-tuned only 45.7% of trainable parameters, reducing end-point error in disparity estimation to 1.85 pixels. Wang et al. introduced the Plant Stereo dataset, providing a high-quality benchmark for plant 3D reconstruction [36]. Ajagekar et al. integrated vision foundation models such as Segment Anything with RGB-D cameras to build a phenotyping system for densely planted lettuce, enabling non-destructive and continuous monitoring of plant height, area, and volume at 10-min intervals, with correlation coefficients exceeding 0.94 compared to manual measurements [37].

This review provides a comprehensive narrative overview of advances in binocular stereo vision for crop phenotyping and smart agriculture, focusing on representative studies published between 2015 and 2025. This time window was selected because deep learning achieved a major breakthrough in stereo matching around 2015. A structured literature search was conducted using databases including Web of Science, Scopus, and ScienceDirect, with keywords covering the intersection of binocular stereo vision, stereo matching, plant phenotyping, crops, and precision agriculture. Relevant studies were identified through the iterative screening of titles, abstracts, and full texts, with priority given to methodologically innovative works and those reporting quantitative performance evaluations. While we aimed for broad coverage across application domains, we acknowledge that the review is narrative in nature and does not follow formal systematic review protocols (e.g., PRISMA). Given the interdisciplinary nature of the field and the diversity of agricultural scenarios, research designs, and evaluation metrics, we consider this narrative approach most suitable for synthesizing current progress and identifying future research priorities.

Overall, binocular stereo vision has evolved from early proof-of-concept studies toward large-scale application-oriented research in agriculture, exhibiting trends toward higher accuracy, real-time processing, lightweight deployment, and improved generalization. However, a significant gap remains between algorithmic performance and large-scale field deployment, particularly in terms of robustness in complex environments, cross-domain generalization, and the lack of standardized evaluation frameworks. Therefore, a systematic review of existing studies is essential for further advancing this field.

2. Fundamentals of Stereo Vision and Adaptation to Agricultural Environments

Stereo vision reconstructs three-dimensional spatial structures from two-dimensional images by simulating the human binocular vision mechanism [38,39]. It serves as a key technical approach for object localization and point cloud reconstruction in agricultural scenes. Despite its strong potential in agricultural phenotyping, its practical deployment in complex field environments remains challenging due to high model complexity, limited dataset availability, and the absence of unified evaluation standards. This section systematically analyzes the core components of binocular stereo vision and their suitability for agricultural applications from a system-level perspective [40,41].

2.1. Construction and Computational Pipeline of Binocular Stereo Vision Systems

The construction and computational pipeline of binocular stereo vision systems consist of three tightly coupled stages: camera calibration, epipolar rectification, and outdoor system design [42,43]. These stages jointly determine the final accuracy of depth estimation.

Camera calibration is the foundational step in stereo vision systems, aiming to establish the mapping relationship between pixel coordinates and 3D world coordinates by estimating intrinsic and extrinsic camera parameters [44–46]. The checkerboard-based calibration method proposed by Zhang has been widely adopted in agricultural applications due to its high accuracy and robust implementation [31]. Zou et al. applied Zhang's calibration method with a ZED 2 stereo camera for field wheat point cloud reconstruction, enabling effective 3D reconstruction of crop scenes [6]. Multi-view calibration experiments in leafy vegetable phenotyping have achieved a reprojection error below 0.18 pixels, providing a reliable geometric basis for subsequent stereo matching [32]. However, even with high-precision camera calibration, residual correspondence and disparity-estimation errors may still propagate into depth estimation and affect the final 3D reconstruction accuracy [47]. In agricultural scenes, these errors can be further aggravated by low-texture surfaces, repetitive plant structures, occlusions, and illumination variations [48,49]. In a systematic review of 114 vision-based pose estimation studies, Menolotto et al. emphasized that calibration strategies should be adapted to deployment scenarios: more complex calibration procedures are suitable for indoor fixed setups, whereas outdoor mobile platforms require a balance between calibration efficiency and long-term stability, along with periodic recalibration mechanisms [50].

In calibrated stereo-vision systems, epipolar rectification transforms the correspondence search from a two-dimensional image-space problem into a one-dimensional search along corresponding epipolar lines, thereby reducing the search space and improving matching efficiency [51–53]. This is a general geometric principle of stereo vision rather than an agriculture-specific procedure. The accuracy of rectification nevertheless directly affects disparity estimation, because residual vertical misalignment may introduce correspondence errors and systematic bias in the reconstructed depth. In agricultural applications, where stereo systems are often mounted on UAVs or mobile platforms, extrinsic parameters may drift due to vibration or mechanical shocks, making periodic recalibration essential for long-term operational stability.

Outdoor system design requires careful consideration at the hardware level. Camera selection must balance resolution, frame rate, baseline distance, and field of view [54]. Menolotto et al. systematically compared the suitability of different camera configurations across zootechnical tasks [50]. Side-view RGB cameras are most suitable for gait and lameness analysis, while top-view RGB setups provide an optimal trade-off for group behavior monitoring. In contrast, stereo vision and multi-view systems are more advantageous when absolute geometric accuracy, occlusion handling, or 3D pose reconstruction is required [54].

For UAV-based deployment, Zhang et al. mounted a ZED 2 stereo camera beneath a plant protection drone and employed an electronic three-axis stabilized gimbal to maintain a consistent vertical viewing angle despite flight attitude variations [31]. The theoretical model of UAV forward flight imaging coverage and camera triggering timing proposed in previous research is visualized in Figure 1, which illustrates the geometric relationship among flight height, field of view, image overlap, and camera moving interval.

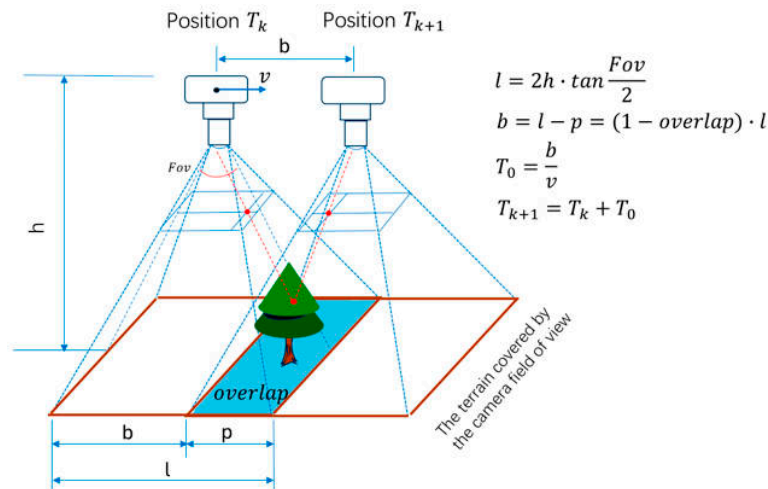


Figure 1. Schematic diagram of UAV sequential imaging geometry [31].

At a flight speed of 2 m/s, the system achieved a maximum relative error of 6.6% in volume estimation for regular cuboid targets. When the speed increased to 6 m/s, the error rose to 7.8%, while the coefficient of variation increased from below 1% to over 3%. These results clearly demonstrate the nonlinear impact of dynamic operating conditions on both measurement accuracy and repeatability.

2.2. Traditional Stereo Matching Algorithms

Stereo matching estimates scene depth by identifying corresponding points between left and right images, computing disparities, and then reconstructing depth via triangulation [55–57]. Traditional stereo matching methods can be broadly categorized into local methods, global methods, and semi-global methods.

Local methods perform correspondence search within a pixel-wise or small window-based neighborhood. They are computationally efficient and highly parallelizable; however, they are prone to mismatches in low-texture, repetitive-texture, and occluded regions. In addition, the use of fixed-size windows limits their ability to balance accuracy and robustness adaptively. Global methods formulate stereo matching as an energy minimization problem and solve it using optimization techniques such as graph cuts and belief propagation. Although they can integrate broader contextual information, their high computational cost makes them unsuitable for real-time applications. Semi-global methods aggregate matching costs along multiple directions, achieving a better trade-off between efficiency and accuracy [58,59].

In agricultural scenarios, the performance of traditional methods is constrained by multiple factors [60]. Yang et al. showed that semi-global methods produced endpoint errors as high as 24.74 pixels across the entire image and 10.87 pixels within leafy vegetable regions [32]. In maize stereo matching studies, Niu et al. observed that SGBM-generated disparity maps were highly fragmented in leaf regions, with blurred edges and numerous holes, whereas deep learning-based methods reduced matching errors by more than one order of magnitude in vegetable regions [35]. These limitations primarily arise from three factors: the lack of distinctive texture information on crop leaf surfaces, illumination variations in field environments that degrade visual consistency, and frequent occlusions caused by overlapping leaves and stems, which result in monocular visibility in certain regions [61–64]. These inherent limitations of traditional methods have directly motivated the adoption of deep learning-based approaches.

2.3. Deep Learning-Based Stereo Matching and Lightweight Edge Deployment

The introduction of deep learning has fundamentally transformed stereo matching, shifting it from hand-crafted feature engineering and heuristic design to a data-driven paradigm that learns feature representations and correspondence relationships automatically [65]. Early deep learning-based stereo matching methods primarily used convolutional neural networks to learn similarity measures or matching costs for stereo image patches [66]. Subsequent studies progressively developed end-to-end architectures that integrate feature extraction, cost-volume construction, cost aggregation, disparity estimation, and refinement within a unified framework [67,68].

Representative supervised learning models have evolved along multiple technical directions. PSMNet introduces spatial pyramid pooling and 3D convolutions to expand the receptive field. GwcNet proposes group-wise correlation to balance efficiency and accuracy. RAFT-Stereo employs a recurrent iterative refinement strategy for disparity optimization. CREStereo adopts a cascaded architecture with adaptive correlation to improve inference speed. In recent years, Transformer-based architectures have also been introduced. Their self-attention mechanism enables global context modeling and mitigates the limited receptive field of convolutional neural networks; however, the associated computational complexity poses significant challenges for deployment on resource-constrained edge devices.

A comparative evaluation of representative stereo matching networks in agricultural leafy vegetable scenarios—including matching accuracy, parameter size, and real-time inference performance on edge platforms—is summarized in Table 1, covering both supervised and self-supervised methods. The endpoint error (EPE) is used to evaluate disparity-estimation accuracy and is defined as the mean absolute difference between the estimated disparity and the ground-truth disparity over all valid pixels: $EPE = \frac{1}{N} \sum_{i=1}^N |\hat{d}_i - d_i^{gt}|$, where N denotes the number of valid pixels, $\hat{d}_i$ is the estimated disparity at pixel i , and d_i^{gt} is the corresponding ground-truth disparity. A lower EPE indicates higher disparity-estimation accuracy.

Table 1. Comparison of the accuracy, model complexity, and inference speed of deep learning-based stereo matching methods in agricultural scenarios.

Model	Architecture	Training Data	Test Scenario	EPE, Full Image (Pixels)	EPE, Target Region (Pixels)	Parameters (M)	FPS
PSMNet	Pyramid Pooling + 3D Convolution	Scene Flow+ fine-tuning	Leaf Vegetable Greenhouse	2.63 pixel	1.14 pixel	5.35 M	1.40
GwcNet	Grouping-related + 3D Convolution	Scene Flow+ fine-tuning	Leaf Vegetable Greenhouse	2.44 pixel	1.03 pixel	6.51 M	7.04
CREStereo	Cascaded Loop + Adaptive Correlation	Scene Flow+ fine-tuning	Leaf Vegetable Greenhouse	3.09 pixel	1.14 pixel	5.43 M	3.52
IGEV-Stereo	ITERATIVE GEOMETRIC CODING + ConvGRU	Scene Flow+ fine-tuning	Leaf Vegetable Greenhouse	2.43 pixel	1.02 pixel	12.59 M	4.71

LVSTereo (Ours)	Multi-scale deformable convolution + composite cost function + 3D depth-separable convolution	3DGS generated (8060 pairs)	Leaf Vegetable Greenhouse	2.32 pixel	0.91 pixel	13.59 M	9.76
LVSTereo (Mip-NeRF supervised)	Multi-scale deformable convolution + composite cost function + 3D depth-separable convolution	Mip-NeRF Generation	Leaf Vegetable Greenhouse	2.84 pixel	1.27 pixel	13.59 M	9.77
PASMN et	Self-supervised learning + parallel attention mechanism	Self-supervised (no true values)	Leaf Vegetable Greenhouse	5.49 pixel	1.52 pixel	7.81 M	23.94
ITSA (transfer learning)	Domain-generalized information theory	Scene Flow pre-training	Leaf Vegetable Greenhouse	4.90 pixel	1.32 pixel	5.22 M	23.18

In agricultural applications, Yang et al. proposed the LVStereo network for leafy vegetable phenotyping, supervised by 3D Gaussian splatting (3DGS) [32]. Multi-view images were first rendered into stereo training data using 3DGS, resulting in a leafy vegetable dataset containing 8060 stereo image pairs. The method introduces a multi-scale deformable convolution-based feature extraction network and a hybrid cost volume design. LVStereo achieved an endpoint error (EPE) of 2.32 pixels on the full image and 0.91 pixels within leafy regions, reaching sub-pixel accuracy for the first time on this dataset and substantially outperforming traditional methods, which reported errors of up to 24.74 pixels.

Similarly, a 2DGS-supervised framework has been developed for maize phenotyping [35]. By leveraging a 2D Gaussian splatting-driven stereo matching pipeline, field-scale 3D phenotyping of maize plants has been enabled. The mean absolute percentage errors (MAPE) for plant height, stem diameter, and leaf area were 3.82, 3.43, and 7.74%, respectively, with all coefficients of determination exceeding 0.95.

In addition, YOLOv11n has been combined with binocular stereo vision for *Camellia oleifera* fruit harvesting [16]. Depth information obtained via stereo matching was converted into 3D coordinates to support vision-based servo control for autonomous picking. The average localization errors along the three spatial axes were 1.70 mm, 1.34 mm, and 1.27 mm, respectively, with a detection time of 58.21 ms, demonstrating the feasibility of deep learning-based stereo matching for real-time agricultural robotic operations.

However, although supervised learning methods perform well on specific benchmark datasets, their direct deployment in agricultural scenarios often suffers from a pronounced “domain shift” problem. Most training data are derived from urban road scenes or indoor environments, which differ substantially from farmland conditions in terms of illumination, texture distribution, and object morphology. To address this issue, a range of domain adaptation strategies have been explored. Self-supervised learning leverages geometric consistency between stereo image pairs as a supervisory signal, eliminating the need for ground-truth annotations [69,70]. Domain adaptation methods typically rely on pretraining on synthetic data followed by fine-tuning in the target domain, or design domain-invariant feature extractors to reduce the gap between source and target distributions [71–73]. Methods such as AdaStereo have made notable progress through output-space self-supervision and feature-space adversarial learning. Nevertheless, the diversity

of agricultural environments—including variations in crop types, growth stages, illumination, and weather conditions—makes it difficult for any single adaptation strategy to achieve comprehensive generalization.

In recent years, the emergence of vision foundation models has provided new opportunities for improving stereo matching generalization. Foundation Stereo, proposed by NVIDIA in 2025, is pretrained on a large-scale synthetic dataset containing one million high-fidelity stereo image pairs, aiming to achieve a strong zero-shot generalization capability. The model adopts a hybrid architecture, incorporating Depth Anything V2 as the core feature extractor to enhance geometric understanding, and introduces attention-based cost aggregation along with ConvGRU-based iterative refinement [74,75]. As noted in the review by Menolotto et al., vision foundation models demonstrate significant potential in agricultural vision tasks; however, model selection should follow a “task-camera-keypoint alignment” principle rather than relying on a single model family [50]. In addition, limited cross-variety and cross-farm data remain a bottleneck.

Lightweight design and edge deployment are essential for translating deep learning-based stereo matching into practical agricultural applications. Agricultural machinery and UAV platforms are typically constrained by limited computational resources, power consumption, and thermal budgets, which further restrict model complexity. Wang et al. proposed a lightweight cattle body measurement method based on YOLOv8n-pose, achieving a 28.3 FPS inference speed on the Jetson Xavier NX platform [76]. A YOLO11-SMMA model has been improved, reducing the parameter count to 1.58 M and achieving a runtime of 58.21 ms on embedded devices, thereby meeting real-time harvesting requirements [16]. However, lightweight design often comes at the cost of reduced accuracy, and achieving an optimal balance between computational efficiency and perceptual performance remains an open problem. For computation-intensive tasks such as stereo matching, model compression and efficient architecture design are key research directions, but the high complexity of agricultural environments imposes strict requirements on robustness; excessive model simplification may lead to severe performance degradation under challenging conditions [77–79].

2.4. Agriculture-Specific Challenges and Adaptation Strategies

Agricultural open-field environments differ fundamentally from structured indoor scenes and standardized autonomous driving roads. Multiple adverse factors coexist, forming major barriers to deploying stereo vision systems from laboratory settings to real farmland [80]. These challenges can be analyzed from five dimensions: illumination, texture, dynamics, occlusion, and environmental interference.

Strong illumination and rapid lighting variations represent one of the most prominent challenges. Field illumination can change by orders of magnitude within short periods, and specular reflections as well as shadows caused by direct sunlight significantly disrupt stereo matching cost computation. Zou et al. explicitly reported that strong sunlight exposure on wheat ears and camera sensors significantly degrades the depth accuracy of the ZED2 system [6]. Zhang et al. further confirmed in UAV-based spraying experiments that outdoor illumination variability directly affects the stability of stereo ranging [31]. Mitigation strategies include hardware-level solutions such as high dynamic range (HDR) cameras, polarization filters, and active illumination, as well as algorithmic approaches based on data augmentation to improve robustness to lighting variations.

Repetitive and low-texture surfaces are inherent characteristics of crop scenes. Most crop leaves exhibit smooth surfaces with limited distinctive features, making them among the most challenging cases for stereo matching. Yang et al. observed that low-texture regions in leafy vegetables led to frequent matching failures in traditional algorithms, while Zou et al. reported increased ambiguity in wheat due to similar textures between ears and

leaves [6,32]. Deep learning methods can partially alleviate this issue by learning high-level semantic representations; however, they depend heavily on large-scale annotated datasets, which remain scarce in agricultural domains.

Dynamic disturbances, including wind-induced leaf motion and platform vibration, further degrade performance. Zhang et al. experimentally showed that increasing UAV flight speed from 2 m/s to 6 m/s raises the relative error in regular object volume estimation from 2.26% to 7.8%, while the coefficient of variation increases from below 1% to over 3%, clearly revealing a nonlinear relationship between motion conditions and measurement accuracy [31]. Motion-related disturbances can be reduced at the hardware level through stabilized camera mounting, gimbal systems, and appropriate image-acquisition settings [81]. At the algorithmic level, temporal information from adjacent frames and motion-aware deblurring methods can be used to improve image consistency under camera or object motion [82].

Occlusion is particularly severe in dense planting systems, where overlapping leaves and stems result in large regions visible only from a single viewpoint, preventing reliable stereo correspondence. Menolotto et al. identified occlusion as a major bottleneck in multi-object and multi-animal pose estimation tasks [50]. Common strategies for occlusion handling include multi-view fusion to increase observational coverage, temporal modeling to exploit growth continuity, and learning-based approaches that explicitly model occlusion patterns to improve matching robustness [83]. Environmental interference includes contamination from dust, dew, insects, and other factors affecting either the camera lens or object appearance [84,85]. Zhang et al. noted that pesticide drift during UAV spraying may adhere to the camera lens and degrade image quality. In addition, temperature and humidity variations can affect sensor performance and calibration stability [31].

From an algorithmic perspective, mitigation strategies can be summarized into four categories: data augmentation, which improves robustness through photometric and geometric transformations; multimodal fusion, which integrates RGB with depth, infrared, or thermal modalities to enhance resilience; temporal modeling, which leverages temporal consistency constraints in video sequences to refine single-frame predictions; and domain adaptation with self-supervised learning, which reduces the discrepancy between training and deployment environments.

Overall, the complexity of agricultural environments requires the systematic co-design of hardware configuration, algorithm selection, and deployment strategies for stereo vision systems. As emphasized by Menolotto et al., vision system design should be guided by the underlying biological or agricultural problem, determining camera geometry, key point definition, and model selection accordingly, rather than merely pursuing the most advanced sensor technologies [50]. As consistently highlighted in the literature, although binocular stereo vision achieves high accuracy in agricultural localization, measurement, and recognition tasks, it still faces challenges such as model complexity, limited scene generalization, insufficient datasets, and the absence of standardized evaluation protocols. Future development therefore requires not only continued algorithmic improvement but also parallel advances in dataset construction and evaluation frameworks. This represents a system-level engineering challenge rather than a problem that can be resolved through isolated technical breakthroughs.

3. Binocular Stereo Vision–Driven Intelligent Applications in Agriculture

Binocular stereo vision has enabled a wide range of intelligent agricultural applications due to its ability to acquire three-dimensional spatial information at relatively low hardware cost [86–89]. As highlighted in the systematic review by Menolotto et al., stereo

vision, RGB-D, and multi-view systems are particularly suitable for tasks requiring absolute geometric accuracy, occlusion handling, or 3D pose reconstruction [50]. Following a hierarchical “crop–environment–equipment–livestock” framework, this section reviews key application scenarios, including high-throughput crop phenotyping, fruit localization and harvesting, weed detection and precision spraying, autonomous navigation of agricultural machinery, and livestock condition assessment, together with their technical approaches and performance boundaries [90–92].

3.1. High-Throughput Crop 3D Phenotyping and Growth Monitoring

Crop phenotypic information constitutes a fundamental data basis for high-throughput breeding and precision crop management. Traditional manual measurements are subjective and low-throughput, making them insufficient to meet the demand for large-scale and high-frequency phenotypic data acquisition in modern breeding programs [93–95]. Owing to its non-contact operation, high efficiency, and capability for 3D reconstruction, binocular stereo vision has become an important tool for crop phenotyping.

In leafy vegetable phenotyping, Yang et al. proposed a stereo matching network named LVStereo, supervised by 3D Gaussian splatting (3DGS), for measuring plant height, leaf number, and leaf area in crops such as lettuce [32]. In this study, multi-view images were rendered into stereo training data using 3DGS, resulting in a leafy vegetable dataset containing 8060 stereo image pairs. A multi-scale deformable convolution-based feature extraction network was further designed. Experimental results show that LVStereo achieved an endpoint error of 2.32 pixels over the full image and 0.91 pixels within leafy regions. For phenotypic parameter estimation, the coefficients of determination (R^2) between predicted and manually measured values reached 0.9946 for plant height, 0.9577 for leaf number, and 0.9775 for leaf area, respectively [96–98].

Ma et al. developed a track-driven greenhouse high-throughput phenotyping platform and employed a multi-view stereo vision approach to acquire time-series point clouds of potted lettuce [99]. Through temporal point cloud registration, individual plant segmentation was achieved, with a vegetation-to-pot segmentation accuracy of 98.81%. The estimated plant height and canopy width achieved mean R^2 values of 0.79 and 0.60, respectively, when compared with ground-truth measurements. Nevertheless, conventional multi-view stereo pipelines fail to handle motion blur and inconsistent illumination in actual greenhouse acquisition. To address this limitation, Yang et al. designed LV-3DGS, a blur-robust 3D Gaussian splatting reconstruction framework, whose detailed workflow is visualized in Figure 2.

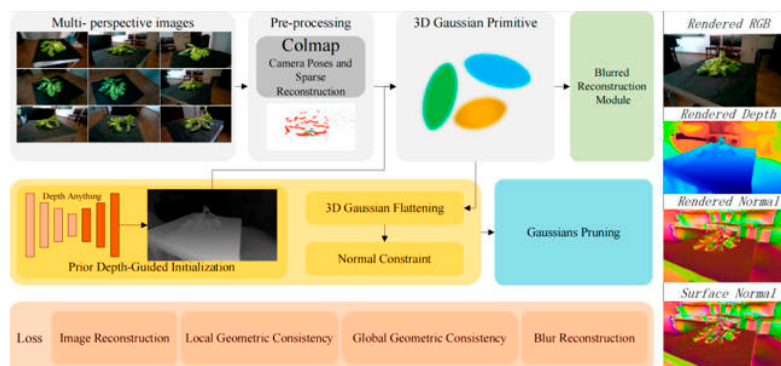


Figure 2. Overall framework for 3DGS-supervised three-dimensional matching and phenotypic measurement of leaf vegetables [32].

In field crop phenotyping, Niu et al. proposed the MaizeStereo framework, which enables 3D maize phenotyping through binocular stereo matching driven by 2D Gaussian splatting (2DGS) [35]. In this study, multi-view images were reconstructed using 2DGS to generate stereo training data, and a stereo foundation model was further fine-tuned via domain adaptation to achieve organ-level phenotypic parameter extraction. Experimental results show that the mean absolute percentage errors (MAPEs) for plant height, stem diameter, and leaf area were 3.822%, 3.432%, and 7.741%, respectively, with corresponding coefficients of determination (R^2) of 0.994, 0.952, and 0.986. These results validate the feasibility of accurate maize phenotyping under complex field conditions.

For rapeseed and other Brassicaceae crops, Wu et al. developed a truss-based field phenotyping platform integrated with BeiDou georeferencing, combining RGB, LiDAR, and multispectral sensors to enable in situ monitoring across the full growth cycle [16]. The platform employs YOLOv11n for plant center localization, achieving an average localization accuracy of 88.2% with a positioning error of 9.3 pixels (approximately 0.011 m) [100,101]. Plant height extracted from point clouds showed strong agreement with manual measurements ($R^2 = 0.9498$, RMSE = 0.0217 m). The processing time for a single plot (30 rapeseed plants, approximately 80,000 points) was about 4 s. In addition, multispectral–RGB fusion was used to monitor frost stress responses in rapeseed, demonstrating the comprehensive potential of stereo vision in multi-growth-stage and multimodal crop monitoring. Zhou et al. systematically investigated crop canopy dynamics from the perspective of time-series point clouds [17]. They constructed a spatiotemporal dataset comprising four crops and eleven varieties and proposed a skeleton-based method for organ-level instance segmentation and temporal analysis.

A systematic evaluation was conducted to quantify the differences in accuracy among various 3D sensing modalities, including binocular stereo vision, multi-view stereo (MVS), RGB-D, LiDAR, and time-of-flight (ToF) systems, across different crop phenotyping tasks. Comparative experimental results involving multiple crops and phenotypic metrics are summarized in Table 2. Using the proposed temporal leaf matching method, the matching accuracy exceeded 0.823 across all species. Furthermore, the study integrates phenotypic tracking results with a canopy photosynthesis model to derive key temporal photosynthetic parameters, including photosynthetic rate, light absorption, and light use efficiency. This enables a complete analytical pipeline ranging from organ-level growth tracking to the evaluation of photosynthetic contribution, providing a quantitative framework for understanding crop growth strategies and light adaptation mechanisms.

Table 2. Comparison of phenotypic measurement accuracy across different crops (multi-method comparison).

Crop Type	Measurement Indicator	Methods for Obtaining Three-Dimensional Information	R^2	RMSE	Experiment Condition
Leafy vegetables (lettuce, etc.)	Plant height	Binocular vision (LVSTereo)	0.9946	0.38 cm	The controlled environment of a greenhouse
Leafy vegetables (lettuce, etc.)	Number of sheets	Binocular vision (LVSTereo)	0.9577	0.96 leaves/plant	The controlled environment of a greenhouse
Leafy vegetables (lettuce, etc.)	Leaf area	Binocular vision (LVSTereo)	0.9775	21.23 cm ²	The controlled environment of a greenhouse
Corn	Plant height	Multi-view Stereoscopic Vision (MVS)	0.998	2.96 cm	Controlled environment in the laboratory

Corn	Leaf area	Multi-view Stereoscopic Vision (MVS)	0.930	75.03 cm ²	Controlled environment in the laboratory
Romaine lettuce	Plant height	RGB-D (multiple views)	0.820	6.20 cm	Greenhouse
Romaine lettuce	Number of sheets	Multi-view Stereoscopic Vision (MVS)	0.979	0.40 leaves /plant	Greenhouse
Romaine lettuce	Leaf area	Multi-view Stereoscopic Vision (MVS)	0.856	1.89 cm ²	Greenhouse
Cucumber seedling	Plant height	Kinect v2 (ToF)	0.982	2.30 cm	Greenhouse
Seedling	Plant height	LiDAR	0.980	2.30 cm	Laboratory

Overall, the application of binocular stereo vision in crop phenotyping is evolving from single-instance static measurements toward time-series dynamic monitoring, and from single morphological descriptors toward multi-organ and multimodal integrated analysis [102–104]. However, three major challenges remain. First, the influence of illumination variation and wind-induced motion on point cloud quality in open-field environments has not been fundamentally resolved. Second, the transformation from point clouds to biologically meaningful phenotypic traits still rely heavily on hand-crafted rules. Third, the generalization capability of models across cultivars and growth stages requires systematic validation. Future research should focus on constructing large-scale time-series point cloud datasets covering a broader range of species and ecological regions, as well as developing end-to-end models for automatic phenotypic trait extraction.

3.2. Fruit Stereo Localization and Autonomous Harvesting in Unstructured Environments

Fruit harvesting is one of the most labor-intensive and automation-demanding processes in agricultural production. Binocular stereo vision provides 3D spatial coordinates of fruits, enabling robotic systems to establish a complete perception loop of “perception–localization–grasping [105,106]”. Unlike 2D detection, which only provides pixel-level localization, stereo vision enables depth perception, allowing robots to accurately plan grasping trajectories in unstructured orchard environments. In camellia fruit harvesting, Zhou et al. combined the YOLOv8x object detection network with binocular stereo vision to achieve 3D localization of picking points [17]. A knowledge transfer strategy was adopted, where the model was pretrained on an apple dataset and then transferred to camellia fruit detection via few-shot learning. With only 1012 annotated samples, YOLOv8x achieved an mAP50 of 0.96. The 3D coordinate estimation error was below 2.1 cm along all three axes. Building on this, Wu et al. proposed a lightweight YOLO11-SMMA model, reducing the parameter size to 1.58 M [16]. On an embedded platform, the inference time was 58.21 ms, and the average localization errors along the X, Y, and Z axes were 1.70 mm, 1.34 mm, and 1.27 mm, respectively. Integrated with a six-degree-of-freedom robotic arm, the system enabled vision-based servo control for automated camellia fruit harvesting, demonstrating the engineering feasibility of lightweight stereo vision solutions in agricultural robotics [107–109].

For kiwifruit harvesting scenarios characterized by dense fruit clusters and severe branch occlusion, stereo vision combined with deep learning-based matching networks has been used to improve the 3D localization of calyx or fruit positions. To further enhance detection and pose estimation under occlusion, multi-view vision information has been fused with LiDAR point clouds. These methods typically use stereo disparity maps or

depth maps to guide noise suppression in LiDAR point clouds, effectively removing spurious points caused by occlusion from branches and leaves. As a result, they significantly improve detection accuracy for occluded fruits. Experiments in real orchard environments show that such multi-sensor fusion approaches outperform conventional pipelines in both localization accuracy and pose estimation while also substantially the reducing filtering time.

Overall, research on fruit localization and autonomous harvesting exhibits three clear trends: a shift from 2D detection to 3D localization, from heavy models to lightweight architectures, and from single-sensor systems to multimodal fusion perception [110,111]. Nevertheless, harvesting success rates remain constrained by three core challenges: accurate reconstruction of fruit geometry under severe occlusion, stability of depth estimation under varying illumination conditions, and end-to-end latency control from perception to grasping. Addressing these challenges requires improvements in stereo matching robustness under occlusion and illumination changes, as well as further acceleration of vision-control closed-loop systems.

3.3. Stereo-Based Weed Recognition and Precision Site-Specific Spraying

Weed competition is a major factor contributing to crop yield loss, potentially reducing yields by up to 34% if not properly managed [112–114]. Conventional uniform herbicide application is not only inefficient in chemical usage, but also leads to severe environmental pollution and herbicide resistance. Binocular stereo vision provides a key perception foundation for precise site-specific spraying by enabling 3D discrimination between crops and weeds [115,116].

From a technical perspective, 3D perception in weed management is mainly utilized in two ways. First, it enhances crop–weed separation by leveraging spatial structure: crops typically exhibit regular row patterns, whereas weeds are randomly distributed [117,118]. Depth and spatial positioning information provided by stereo vision significantly improves this distinction. Second, 3D reconstruction enables the accurate spatial localization of weeds, guiding variable-rate sprayers or laser-based weeding systems for targeted removal.

In precision spraying systems, a WEED-YOLOv10-based targeted spraying platform has been validated in maize fields. The system performs real-time weed localization and controls spraying units for precise application. Field experiments show a targeting accuracy of 93.7%, spray coverage of 90.5%, a targeting deviation of 1.45 cm, and a detection speed of 20.1 cm/s. The variable-rate application system reduces overall application error to within 3% and achieves pesticide savings of 40–70%.

For in-row weed recognition in robotic weeding systems, binocular stereo vision is used to distinguish weeds from crop plants based on 3D spatial information. A stereo vision-based in-row weed identification method has been proposed for the effective detection of early-stage maize weeds. In orchard environments, a single-row tracking robot integrating stereo vision and LiDAR has also been developed, where deep learning-driven 3D perception triggers lateral weeding mechanisms, improving operational effectiveness in orchard inter-row management.

At a broader level, 3D sensing technologies are reshaping precision weed management. Existing reviews indicate that LiDAR–RGB fusion provides the most reliable outdoor 3D weed detection performance in dense canopies, while the combination of SfM and MVS achieves the lowest reconstruction error in dense planting scenarios [119,120]. However, these studies also highlight a critical limitation: model generalization across fields remains weak due to variations in crop species, soil conditions, canopy occlusion, and weed growth stages. Limited labeled datasets and high computational costs further

restrict field deployment. This limitation equally applies to binocular stereo vision systems in weed management, where a single model struggles to generalize across diverse crops, growth stages, and ecological regions.

Overall, stereo-based weed identification and precision site-specific spraying have progressed from proof-of-concept studies to field-scale validation, establishing an initial technical foundation for practical deployment. However, large-scale commercialization remains limited by several key bottlenecks. First, the 3D discrimination accuracy between crops and weeds is still insufficient for precision weeding in high-value crops. Second, the trade-off between real-time processing capability and field operation speed has not been fully resolved. Third, model generalization across regions and seasons remains inadequate and requires systematic improvement.

3.4. Vision-Based Navigation of Agricultural Machinery and 3D Evaluation of Operational Quality

Autonomous navigation of agricultural machinery is a core component of intelligent agriculture and directly affects operational accuracy, efficiency, and safety. Traditional navigation systems primarily rely on Global Navigation Satellite Systems (GNSS) [121]. However, in high-density environments such as orchards, signal degradation and multipath effects can severely reduce GNSS reliability. As a self-contained perception modality, binocular stereo vision provides an important complement for continuous navigation in GNSS-denied environments.

In orchard sprayer navigation, researchers have developed deep learning-based stereo vision navigation systems that integrate object detection models such as EfficientDet or YOLOv7 with stereo localization. These systems provide inter-row positioning and navigation guidance for air-assisted orchard sprayers. In field tests conducted on a four-row orchard road, the EfficientDet-based navigation algorithm achieved minimum average RMSE values of 0.24 m, 0.20 m, and 0.31 m at travel speeds of 1, 1.5, and 2 km/h, respectively. The system also integrates RGB-D-based collision avoidance modules, demonstrating operational safety under real-world conditions.

For autonomous navigation of tractor-trailer systems, a two-stage stereo vision-based 3D obstacle detection framework combined with Frenet-optimal trajectory planning has been proposed [122–124]. The system extracts 3D obstacle information from stereo vision and performs trajectory optimization to enable autonomous path tracking of the trailer system. In rice transplanter navigation, a hybrid system integrating machine vision and GNSS has been developed for boundary detection and positioning, enabling continuous autonomous operation and automatic headland turning.

In terms of operational quality assessment, binocular stereo vision has also been applied to real-time canopy volume estimation in spraying operations. Zhang et al. mounted a ZED 2 stereo camera beneath a plant protection UAV to perform online estimation of fruit tree canopy volume during flight [31]. By combining canopy segmentation and sub-grid-based volume computation, the system enables real-time canopy volume extraction. At a flight speed of 2 m/s, the system-maintained volume estimation errors within 10% for regular targets, providing real-time decision support for variable-rate spraying. Further analysis showed that when flight speed increased from 2 m/s to 6 m/s, both measurement error and variability increased significantly, with the coefficient of variation rising from below 1% to over 3%, revealing a nonlinear relationship between dynamic conditions and 3D measurement accuracy.

Overall, vision-based navigation of agricultural machinery is evolving from single-row detection toward multi-task, multi-sensor integrated perception systems. Binocular stereo vision is shifting from an auxiliary positioning role to a core sensing component. Nevertheless, a gap remains between current navigation accuracy (decimeter-level) and

the requirements of precision agricultural operations (centimeter-level). In addition, system reliability under adverse weather conditions such as rain, fog, and dust still requires further validation. Future research should focus on multi-source fusion navigation architectures integrating vision, inertial sensing, and GNSS, as well as robust perception algorithms capable of operating under extreme illumination and weather conditions.

3.5. Stereo-Based Livestock Phenotyping and 3D Behavioral and Body Condition Assessment

Livestock body condition is a fundamental indicator in precision animal husbandry, directly influencing feeding strategies, reproductive management, and disease early warning [125,126]. Traditional manual scoring methods are subjective, labor-intensive, and unsuitable for large-scale, high-frequency monitoring. Binocular stereo vision and 3D point cloud reconstruction provide a non-contact solution for livestock body condition assessment. As noted by Menolotto et al., stereo vision, RGB-D, and multi-view systems are particularly valuable for tasks requiring absolute geometric accuracy, occlusion handling, or 3D pose reconstruction, especially in body measurement and weight estimation [50].

In body weight estimation, binocular stereo vision has been applied to non-contact weighing of various livestock species. A non-contact yak body weight estimation method based on skeletal keypoint localization has been proposed [76]. YOLOv8 was used for yak detection and keypoint extraction (hip, ischial, shoulder, and hoof points), combined with stereo vision to compute body length, diagonal length, and withers height. Field data collected from yaks on the Qinghai–Tibet Plateau were used to train a Gaussian process regression model, achieving a mean absolute error (MAE) of 25.4 kg and an R^2 of 0.72 in cross-validation over 50 yaks. Based on this system, a Raspberry Pi-based intelligent inspection vehicle and a handheld device were further developed to enable real-time non-contact weight estimation.

For pig weight estimation, a LabVIEW-based binocular stereo vision system was applied to analyze body dimensions and live weight of 10 Landrace pigs aged 14 to 25 weeks [127]. Three-dimensional body measurements were extracted using stereo imaging, and regression models were established between body parameters and weight to enable non-contact estimation. A review further compared image-based and point cloud-based approaches for pig weight estimation, highlighting differences in methodology and performance.

In cattle body condition scoring, computer vision models based on top-view images have been developed to automatically segment cattle bodies and detect keypoints associated with body condition assessment. Three-dimensional top-view imaging provides distinct advantages in estimating body condition score, growth development, and morphological traits. For sheep body condition evaluation, Cowie et al. proposed a video-based temperament prediction framework integrating computer vision, behavioral clustering, and machine learning [128]. Using arena test videos from 63 lactating ewes, kinematic features were extracted, and a random forest model achieved a classification accuracy of 77%. For sheep behavior recognition, Xue et al. proposed a DMS-Net dynamic multi-scale multimodal fusion network that integrates visual and skeletal modalities to classify standing, lying, kneeling, and sitting postures [129]. The model achieved an accuracy of 98.89% under complex housing conditions.

In the field of pose estimation and behavioral analysis, the review by Menolotto et al. systematically examined 114 vision-based pose estimation studies [50]. The authors reported that side-view RGB imaging is most suitable for gait and lameness analysis, whereas top-view RGB configurations provide the best overall performance for group behavior monitoring, feeding analysis, tracking, and identity recognition tasks. However, the review also highlights several critical limitations that have constrained progress in this

area, including inconsistent definitions of anatomical landmarks, fragmented datasets, heterogeneous evaluation metrics, and insufficient reporting of runtime performance, calibration requirements, and system-level farm integration. In this context, representative studies applying binocular stereo vision to livestock body weight estimation, pose recognition, and body condition scoring are systematically reviewed. The sensing hardware, algorithmic architectures, and quantitative performance of these approaches are summarized in a unified manner in Table 3.

Table 3. Summary of research on three-dimensional phenotypic acquisition and three-dimensional evaluation of body condition behavior in livestock and poultry.

Types of Live Stock and Poultry	Research Objectives	Three-Dimensional Perception Methods	Core Method	Key Performance Metrics	Performance Data
Yak	Non-contact weight estimation	No model specified for binocular vision)	YOLOv8 skeletal key points (hip/ischium/shoulder/hoof) + Gaussian process regression	MAE/R ²	25.4 kg/0.72
Yak	Non-contact body weight estimation (edge deployment)	Binocular Vision + Raspberry Pi	Smart Inspection Vehicle/Handheld Detection Device	Reasoning Speed (Light-weight/High-weight Variant)	0.78 s/image
Longbai Pig	Non-contact weight estimation	Binocular stereoscopic vision (LabVIEW platform)	Size Dimensionality Regression Model	Suitable object, applicable targets	$n = 10$, age = 14–25 weeks
Cow	Body condition score (BCS)	Top View RGB/3D Image	Key Point Detection + Automatic Scoring	—	The three-dimensional top view offers unique advantages in BCS estimation.
Sheep (lactating ewes)	Temperament prediction and classification	RGB video (GoPro HERO 8, top-down view)	YOLOv8x-seg + BoT-SORT tracking + Kinematic features + Random Forest	Test accuracy (RF model)	77% (Three types of temperament clusters)
Sheep	Posture recognition (standing/lying/kneeling/sitting)	RGB video (livestock pen monitoring)	DMS-Net (Visual-Skeletal Dual-Modality Fusion + MS-GCN)	Identification accuracy rate (complex pen facilities)	98.89%
Sheep	Pose recognition (obstruction robustness)	RGB video (livestock pen monitoring)	DMS-Net (Dual-modal Fusion)	Accuracy rate at 30% bone occlusion	98.56%
Cattle	Pose estimation and limp detection	RGB video (side view)	A Systematic Review of 114 Studies on Visual Pose	Key findings	Side-view RGB is best suited for gait analysis

Overall, livestock 3D phenotyping and body condition assessment are at a critical transition stage, moving from laboratory validation toward on-farm deployment. The advantages of binocular stereo vision in this domain lie in its non-contact operation, non-intrusive nature, and capability for high-frequency monitoring, which are well-aligned with the principles of modern animal welfare-oriented husbandry. However, practical challenges such as occlusion caused by fur, highly variable postures, and severe inter-individual occlusion in group settings impose stringent requirements on the robustness

of stereo matching algorithms for non-rigid targets. Future research should focus on establishing unified keypoint annotation standards across species and breeds, constructing large-scale 3D datasets covering diverse husbandry conditions and body states, and developing lightweight pose estimation models capable of real-time inference on edge devices.

From a broader perspective, applications of binocular stereo vision in agricultural intelligence exhibit a clear evolutionary trajectory from “seeing” to “measuring accurately,” and further toward “understanding”. In crop phenotyping, stereo vision has achieved sub-pixel-level reconstruction of plant 3D structures and is now extending toward time-series dynamic monitoring and photosynthetic function assessment. In fruit harvesting, the technology has progressed from laboratory proof-of-concept to field deployment in orchards, where lightweight design and real-time performance have become central engineering requirements. In weed management, 3D spatial perception provides a foundation for precise variable-rate spraying, although cross-scene generalization remains a major bottleneck. In agricultural machinery navigation, stereo vision is evolving from an auxiliary positioning tool to a core perception modality. In livestock monitoring, non-contact body condition assessment has demonstrated clear application value, yet large-scale commercial deployment remains limited.

4. Datasets, Benchmarks, and Major Challenges

High-quality datasets and standardized evaluation benchmarks are fundamental infrastructure for continuous progress in stereo vision algorithms. In the field of autonomous driving, benchmark datasets such as KITTI, Scene Flow, and Middlebury have directly driven the paradigm shift from traditional stereo methods to deep learning-based approaches. In contrast, agricultural stereo vision research has long lacked comparable data infrastructure, which has become a key bottleneck limiting field development [130,131]. Even when data are available, challenges such as illumination variability in open environments, real-time constraints, multimodal fusion requirements, and the interpretability of safety-critical systems continue to hinder the translation from algorithmic research to field deployment. This section therefore analyzes these challenges in a structured manner, starting from data foundations.

4.1. Public Agricultural Stereo/RGB-D Datasets and Standardized Benchmarks

The development of agricultural stereo vision datasets has consistently lagged behind algorithmic advances, primarily due to several structural constraints [132–134]. Unlike autonomous driving datasets, which can be collected continuously from vehicles in motion, agricultural data acquisition is constrained by crop growth cycles, which typically span several months or even a full year; missing a collection window often requires waiting for the next season. Annotation is also significantly more expensive: precise delineation of plant organs is far more time-consuming than bounding-box annotation for vehicles or pedestrians, while obtaining ground-truth disparity at sub-pixel accuracy is practically infeasible through manual labeling. Together, these factors result in limited scale, diversity, and annotation precision in agricultural stereo datasets.

Recently, researchers have begun to systematically address this gap. The PlantStereo dataset, which contains 812 stereo image pairs covering spinach, tomato, pepper, and pumpkin, has been proposed [36]. Ground-truth disparities were generated using a semi-automatic pipeline based on depth images, enabling relatively high annotation accuracy. However, the dataset size remains limited for deep network training. In comparison, the Scene Flow dataset in autonomous driving contains more than 39,000 synthetic stereo pairs.

As highlighted by Menolotto et al. in a systematic review of 114 vision-based pose estimation studies, inconsistent landmark definitions, fragmented datasets, and heterogeneous evaluation metrics represent key barriers to progress in the field [50]. A similar issue exists in stereo matching, where evaluation protocols are highly fragmented: some studies report endpoint error (EPE), others use D1 error or Bad-1/Bad-3/Bad-5 metrics, while some adopt PCK or OKS, making cross-study comparisons difficult [135,136]. Given the inherent diversity of agricultural scenarios, it is more practical to develop domain-specific benchmarks—such as orchard fruit detection benchmarks, leafy vegetable phenotyping benchmarks, or field crop height estimation benchmarks—rather than pursuing a single unified agricultural stereo matching benchmark.

To clearly illustrate the large-scale gap between agricultural stereo datasets and those in autonomous driving and general vision benchmarks, a comparative summary of dataset size, annotation types, and acquisition methods is provided in Table 4.

Table 4. Comparison of scale between the agricultural 3d dataset and autonomous/general purpose datasets.

Dataset	Industry Sector	Image Logarithmic Scale	Annotation Type	Collection Method	Gapin Scale with Agricultural Datasets
Scene Flow	Autonomous driving/general purpose	>39,000 pairs of composite stereoscopic images	Parallax-based optical flow scene stream	Synthesis (animation rendering)	48 times for PlantStereo (812 pairs)
KITTI 2015	Autonomous driving	200 pairs (actual labeling)	Parallax (LiDAR sparse ground truth, $\approx 19\%$ density)	On-vehicle field collection (Karlsruhe, Germany)	Its scale is smaller than that of the agricultural dataset, but its evaluation system is mature.
Middlebury 2014	General (indoor)	33 pairs	Sub-pixel level parallax density	Structural light camera (for laboratory use)	Small in size but high in accuracy, serving as a benchmark for algorithm validation.
ETH3D	General-purpose (indoor/outdoor)	47 pairs	Parallax density	Laserscanner	Covers multiple scenarios, but with limited scale
Cityscapes	Autonomous driving (urban environment)	3475 pairs	Parallax (generated by SGM, $\approx 38\%$ density)	The vehicle-mounted binocular camera captures images.	Large in size, but labeled as algorithm-generated rather than actual values
PlantStereo	Agricultural (plants)	812 pairs	Sub-pixel level parallax density	Semi-automatic deep image generation	The first specialized three-dimensional dataset for agriculture
Leafy Vegetable Dataset	Agricultural (leafy vegetables)	8060 pairs	Sub-pixel level parallax density	3DGS rendering generated (36 scenes)	The largest agricultural-specific 3D dataset available today

4.2. Robustness Under Variable Illumination and Adverse Weather Conditions

Illumination conditions in agricultural environments exhibit an extremely wide dynamic range, spanning from tens of thousands of lux under direct midday sunlight to only a few hundred lux under overcast conditions. Such variations can occur within short time intervals, resulting in changes of up to two orders of magnitude in lighting intensity. Overexposed regions lead to the loss of texture information, while shadowed regions significantly reduce the discriminative power of matching features. In addition, changes in illumination direction alter the specular reflection characteristics of leaf surfaces, causing the same object to exhibit substantially different appearances across time and viewpoints.

Zou et al. explicitly reported a significant degradation in depth estimation accuracy of the ZED 2 camera under strong sunlight in field wheat experiments [6]. Similarly, Yang et al. observed that even under controlled greenhouse conditions, illumination variations could lead to a large number of matching failures in traditional stereo matching algorithms [32]. To tackle the image blur problem induced by unstable lighting and motion during field data acquisition, we designed a dedicated deblurring module, and its qualitative comparison results are illustrated in Figure 3.

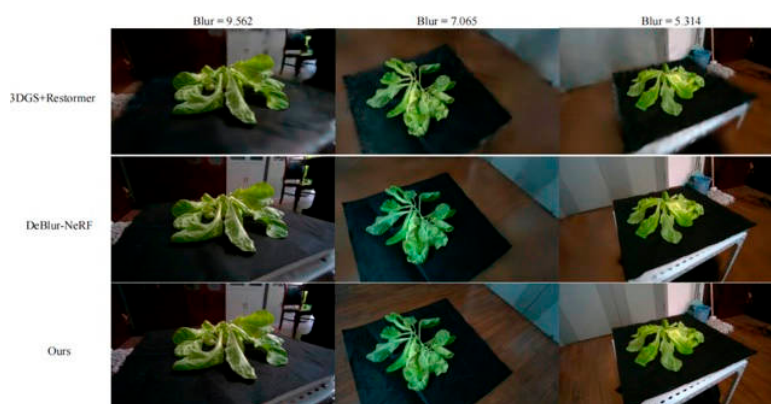


Figure 3. The visualization results of the deblurring module [32].

Adverse weather conditions further exacerbate illumination-related challenges. In rainy conditions, water droplets adhering to the camera lens can obstruct the field of view and introduce refractive distortions [137–139]. In foggy environments, aerosol scattering leads to a significant reduction in image contrast. In dusty conditions, particulate contamination on the lens alters the light propagation path and degrades image quality. However, existing agricultural stereo datasets rarely include such extreme conditions, resulting in a lack of systematic evaluation of model performance under adverse weather.

From an algorithmic perspective, current mitigation strategies mainly focus on three directions. First, data augmentation simulates diverse illumination conditions by applying transformations in brightness, contrast, and related photometric properties to training samples. Second, domain adaptation methods aim to reduce feature distribution discrepancies across illumination conditions using adversarial training or self-supervised learning. Third, multimodal fusion combines RGB with infrared or thermal imaging, leveraging their complementary responses to illumination variations to improve robustness. Pham et al. proposed the CattleFever system, which integrates RGB and thermal imaging for facial key point localization in cattle [103]. The system achieved an average precision (AP) of 94.37% using RGB images and 73.16% using thermal images, demonstrating the inherent robustness of thermal imaging to illumination changes. Nevertheless, each of these strategies has inherent limitations. Data augmentation cannot fully cover the space

of possible illumination variations; domain adaptation relies on prior access to target-domain data; and multimodal fusion increases hardware cost and system complexity. To date, no single approach can reliably handle all illumination variation scenarios. A comprehensive comparison of stereo vision performance degradation under varying illumination conditions, weather states, and crop growth stages across multiple field scenarios—including wheat, kiwifruit, maize, and livestock environments—is summarized in Table 5.

Table 5. Robustness validation of the binocular vision system under varying lighting and weather conditions.

Research Scenario	Weather Conditions for Solar Radiation	Sensing Device	Performance Index	Performance Under Normal Conditions	Performance Under Harsh Conditions	Degree of Performance Degradation
Wheat density estimation (field)	Direct intense light	ZED2 Binocular Camera	Depth measurement accuracy	Effective under normal light conditions	Precision decreases significantly under strong light.	Direct exposure to intense light has a significant impact on both the spikelet and the camera.
Kiwi fruit posture estimation	Different time periods (9–11 AM/12–3 PM/4–7 PM)	ZED 2i + Livox Mid70	Positioning MAE/posture error	9–11AM: 7.26 mm/8.29°	4:00–7:00 PM: 4.63 mm/5.02 (optimal)	Changes in light intensity cause the MAE to fluctuate between 4.63 and 7.87 mm.
Kiwi fruit posture estimation	Different time periods	ZED 2i + Livox Mid70	Attitude error (θ_{err})	9–11AM: 8.29°	12:00–3:00 PM: 8.61	The all-day attitude error fluctuates between 5.02 and 9.95.
Estimation of fever in cattle	Different lighting conditions (RGB vs. thermal imaging)	RGB + Thermal Imaging (ICI FMX 400)	Key point AP	RGB: 94.37%	heat-imaging, thermal imagery, thermal imaging	Thermal imaging is inherently insensitive to variations in light intensity, but exhibits low resolution.
Kiwi Stereo matching	Four scenarios: overexposure, low light, oblique lighting, etc.	GuideNet (Binocular Stereo-scopic Matching)	Feature matching quality	Good matching under normal lighting conditions	Remains effectively matched under overexposure or low light conditions	GuideNet can accurately reconstruct branch structures even under extreme lighting conditions.
Maize stereo matching (field)	Different growth stages (V3 → R3)	Binocular Vision + Maize Stereo	EPE (end point error)	V3–V5: Low Error	V12–R3: Error increases (due to more blades and intensified texture repetition)	Increased structural complexity leads to a rise in EPE.

4.3. Real-Time Constraints and Edge Computing Adaptation

The deployment platforms of agricultural stereo vision systems—such as field robots, unmanned aerial vehicles (UAVs), and vehicle-mounted devices—are typically subject to stringent computational resource constraints [140]. Compared with data-center-level GPUs, edge devices offer significantly lower computational power, limited energy budgets, and restricted thermal dissipation capability. Stereo matching, however, is one of the most computationally intensive tasks in computer vision.

As systematically reviewed by Menolotto et al., several studies have evaluated inference performance on edge platforms [50]. Weng et al. reported a lightweight cattle body measurement method, achieving 28.3 FPS on the Jetson Xavier NX [76]. Wu et al. reduced the parameter size of YOLO11-SMMA to 1.58 M and achieved an inference time of 58.21 ms [16]. On embedded platforms such as the NVIDIA Jetson series, prior studies have also reported goat behavior recognition speeds of approximately 24 FPS. However, it should be emphasized that these real-time performances are mainly achieved in 2D object detection tasks. In contrast, stereo matching requires a dense correspondence search across the disparity dimension, and its computational complexity typically increases linearly with the disparity search range, resulting in substantially higher computational demand than 2D detection under the same input resolution. To enable a direct and transparent comparison of these widely scattered real-time performance figures, we consolidated the key metrics from the literature into Table 6.

Table 6. Comparison of real-time inference performance of representative agricultural vision models on edge devices.

Study/Task	Model/Method	Hardware Platform	Task Type	Inference Speed	Key Notes
Wang et al. (yak weight) [76]	YOLOv8n-pose	Jetson Xavier NX	2D pose estimation + key-point detection	28.3 FPS	Lightweight design for mobile inspection vehicle.
Wu et al. (oil-tea fruit harvesting) [16]	YOLO11-SMMA (1.58 M params)	Embedded device (NVIDIA Jetson series)	2D detection + binocular stereo localization	58.21 ms/frame (~17.2 FPS)	Includes detection and stereo matching; meets real-time harvesting requirements.
Xue et al. (sheep behavior) [129]	DMS-Net (dynamic multi-scale multi-modal)	NVIDIA Jetson series	2D behavior recognition (standing, lying, etc.)	~24 FPS	Fuses vision and skeletal modalities; maintains high frame rate in complex housing conditions.
SOTA stereo matching (e.g., RAFT-Stereo/CREStereo) [75]	Dense stereo matching network	Jetson Orin Nano Super	Binocular dense disparity estimation	~6.99 FPS	High accuracy but computationally intensive; reveals the fundamental bottleneck for pure stereo matching on edge.

It has been demonstrated in yak body weight estimation that excessive model lightweighting can lead to a sharp degradation in performance under complex conditions [76]. In their experiments, a lightweight model required approximately 0.78 s per inference on a Raspberry Pi, while heavier variants required several seconds. The high complexity of agricultural environments imposes strict requirements on robustness, and an optimal trade-off between lightweight design and accuracy has not yet been achieved.

4.4. Multimodal Fusion and Synthetic Data-Driven Augmentation

The limitations of single RGB modality in agricultural scenarios are becoming increasingly evident. Variations in illumination, lack of texture, and severe occlusion make

it difficult for pure vision-based approaches to maintain stable performance under all conditions. Multimodal fusion and synthetic data-driven augmentation are widely regarded as two complementary strategies to address these limitations.

In multimodal fusion, RGB-D cameras simultaneously provide color images and depth information, enabling algorithms to exploit both appearance and geometric cues. As noted by Menolotto et al., RGB-D, stereo vision, and multi-view systems are particularly advantageous for tasks requiring absolute geometric accuracy, occlusion handling, or 3D pose reconstruction [50]. Xue et al. proposed DMS-Net, a vision–skeleton multimodal fusion framework for recognizing four sheep postures, achieving an accuracy of 98.89% in complex housing environments [129]. More advanced fusion strategies include the integration of RGB with LiDAR, thermal imaging, or multispectral/hyperspectral data.

However, multimodal fusion introduces significant system-level engineering challenges, particularly in spatial calibration and temporal synchronization between sensors. Even millimeter-level calibration errors can lead to substantial deviations in 3D geometric reconstruction. In addition, the increased hardware cost and system complexity cannot be neglected, and sensor configuration must be carefully balanced between information gain and implementation cost.

Synthetic data provide an alternative solution to the scarcity of agricultural datasets. Yang et al. generated stereo training data using 3D Gaussian splatting (3DGS) from multi-view images, constructing a leafy vegetable dataset containing 8060 stereo image pairs [32]. Niu et al. adopted a 2D Gaussian splatting (2DGS)-based approach for maize phenotyping, while Lin et al. developed the UE5-Forest dataset entirely within the Unreal Engine 5 simulation environment [35]. The key advantage of synthetic data lies in fully automated, zero-cost annotation, enabling the simultaneous generation of disparity, depth, surface normals, and semantic segmentation labels. However, the primary limitation is the domain gap between synthetic and real-world data, which must be mitigated through domain adaptation techniques.

4.5. Explainability and Safety-Critical System Design

As stereo vision systems transition from laboratory research to real-world agricultural deployment, their role shifts from analytical tools to decision-making components, directly controlling robotic arms for harvesting, spraying nozzles, or autonomous agricultural machinery. This transition makes system explainability and safety unavoidable engineering concerns [141–143].

From the perspective of interpretability, the black-box nature of deep learning models is particularly problematic in agricultural applications [144,145]. When a stereo matching network produces erroneous disparity estimates that lead to failed robotic grasping in orchards, it is necessary to trace the underlying source of failure. However, explainability research in stereo matching remains significantly less developed than in image classification and object detection. Methods such as Grad-CAM provide limited interpretability for dense prediction tasks. As noted by Menolotto et al., interpretability in pose estimation is mainly achieved through skeleton overlay visualization, feature importance analysis, and 3D evidence representation; however, truly post hoc explainable AI remains limited, and explicit user-oriented XAI solutions are still the exception rather than the norm [50].

To mitigate the “black-box” nature of deep learning models in complex orchard environments, enhancing the visual interpretability of fruit feature perception through visualization techniques is essential. It has been observed that lightweight detection models often fail to focus accurately on target fruits under dense occlusion and background interference, leading to semantic drift in feature extraction. Representative detection visuali-

zation results of two mainstream fruit detection algorithms under dense blueberry occlusion scenes from the existing research are displayed in Figure 4, which intuitively reflects the difference in target localization performance between different detectors.

From the perspective of safety-critical system design, the deployment of agricultural stereo vision systems presents unique challenges. Unlike autonomous driving, agricultural scenarios lack standardized safety regulations and technical protocols. The consequences of vision system failure in harvesting robots differ substantially from those in autonomous navigation machines, necessitating dedicated redundancy designs and fail-safe mechanisms [146–148]. At present, safety design in agricultural stereo vision systems remains at an early stage, with most studies focusing on algorithmic accuracy while insufficient attention is paid to system-level reliability and robustness. Future work should establish safety standards and testing protocols tailored to agricultural robotic vision systems, covering key aspects such as sensor fault detection, uncertainty quantification in algorithms, and decision confidence estimation. As emphasized by Menolotto et al., deployment-oriented studies should go beyond reporting nominal frame rates and instead include critical information such as calibration burden, environmental robustness, runtime performance, edge hardware constraints, and integration with farm management systems [50].

Overall, agricultural stereo vision is undergoing a critical transition from an “algorithm-driven” paradigm toward a “data–algorithm–system co-driven” framework. In terms of datasets and benchmarks, resources such as PlantStereo and AppleGrowthVision have partially filled the existing gaps; however, dataset scale, diversity, and standardization remain insufficient. Establishing a benchmark comparable to KITTI in autonomous driving remains an open challenge. In terms of illumination robustness, data augmentation, domain adaptation, and multimodal fusion provide partial solutions, but no method can reliably handle all illumination conditions. In terms of real-time performance, a significant gap remains between edge-device inference speed and real-world operational requirements. For multimodal fusion, combinations such as RGB-D, RGB-LiDAR, and RGB-thermal imaging have demonstrated performance gains, but at the cost of increased system complexity and hardware overhead. In terms of explainability and safety, research is still in its early stages [149,150].



Figure 4. Visual comparison of blueberry fruit detection results. The first row presents raw orchard input images with heavy fruit occlusion; the second and third rows illustrate the prediction outputs generated by YOLOv12m and RT-DETR-v2-X, respectively, where red boxes denote detected blueberry fruits and green boxes mark false-positive or missed detection regions [151].

These challenges are strongly interdependent: insufficient datasets constrain robust algorithm development, computationally intensive models limit edge deployment, and limited deployment experience hinders the establishment of safety standards. Breaking this cycle requires coordinated progress across datasets, algorithms, hardware, and system design rather than isolated advances in any single component.

5. Challenges and Future Perspectives

5.1. Key Findings Across Application Domains

Based on a systematic review of binocular stereo vision fundamentals, algorithmic evolution, and agricultural applications, four key findings can be summarized.

At the technical level, traditional stereo matching methods perform adequately in structured greenhouse environments but degrade significantly in open-field scenarios characterized by weak textures, strong illumination variations, and severe occlusions. In wheat density estimation, Zou et al. reported endpoint errors as high as 24.74 pixels using semi-global matching methods. In contrast, deep learning-based approaches have fundamentally improved performance. For example, Yang et al. achieved sub-pixel accuracy in leafy vegetable phenotyping using the 3DGS-supervised LVStereo network, reducing endpoint error in leaf regions to 0.91 pixels. This gap clearly demonstrates the superiority of data-driven methods in complex biological structures.

At the data infrastructure level, agricultural stereo vision suffers from persistent data scarcity. PlantStereo contains only 812 stereo pairs, whereas Scene Flow in autonomous driving includes more than 39,000 pairs. The release of AppleGrowthVision (9317 stereo

images across six growth stages and 31,084 annotated apple instances) represents a significant milestone in orchard-scale dataset construction. UE5-Forest exemplifies the frontier of synthetic data generation, producing 5520 stereo image pairs with pixel-perfect disparity in Unreal Engine 5. However, as highlighted by Menolotto et al. in a review of 114 studies, inconsistent landmark definitions, fragmented datasets, and heterogeneous evaluation metrics remain major barriers to progress.

At the application level, stereo vision has been widely adopted in crop phenotyping, fruit harvesting, weed control, agricultural machinery navigation, and livestock monitoring. Each application imposes distinct requirements: phenotyping demands sub-millimeter accuracy and temporal consistency; harvesting emphasizes real-time and lightweight computation; weed management requires strong cross-domain generalization; navigation demands robustness; and livestock monitoring involves non-rigid object modeling. As emphasized by Menolotto et al., side-view RGB is most suitable for gait analysis, top-view RGB performs best for group behavior analysis, while stereo vision and multi-view systems are preferred when absolute geometric accuracy or 3D pose recovery is required. This highlights a key principle: performance depends on task–camera–keypoint alignment rather than reliance on a single model family.

At the challenge level, significant gaps remain between current systems and large-scale deployment. No existing method can fully handle illumination variability. Although recent edge systems-on-chip have substantially improved computational capability, real-time performance for computationally intensive stereo-matching networks remains dependent on the specific hardware platform, model complexity, input resolution, and power configuration. For example, a representative state-of-the-art stereo network achieved approximately 6.99 FPS on the Jetson Orin Nano Super. More powerful platforms, such as the Jetson AGX Orin and Jetson Thor, provide greater computational capacity, but they also involve trade-offs in power consumption, cost, and thermal dissipation. Multimodal fusion introduces additional calibration and system complexity, while research on explainability and safety is still in its infancy.

5.2. Practical Roadmap from Research Prototypes to Deployment

Based on these findings, a staged roadmap is proposed for transitioning agricultural stereo vision from research prototypes to real-world deployment.

Stage I (0–3 years): Standardization and infrastructure development.

The most urgent task is to establish standardized benchmarks for agricultural sub-domains. As demonstrated by AppleGrowthVision, benchmark datasets can significantly improve model performance (e.g., +7.69% F1-score for YOLOv8 and +31.06% for Faster R-convolutional neural network). Future efforts should focus on constructing benchmarks for orchard fruit detection, leafy vegetable phenotyping, and field crop height estimation. Synthetic data pipelines such as UE5-Forest provide a scalable solution when real data are difficult to acquire. Standardized reporting guidelines should also be established, including calibration effort, robustness evaluation, runtime, edge-device performance, and system integration metrics.

Stage II (3–5 years): Domain adaptation and lightweight modeling.

Foundation Stereo demonstrates strong zero-shot generalization through large-scale pretraining on one million stereo pairs. Similarly, feature-adaptive strategies leveraging monocular priors have shown promise in reducing the sim-to-real gap. These approaches are particularly critical for agricultural scenarios. Future work should focus on domain-adaptive fine-tuning of foundation models for agriculture, combined with pruning, knowledge distillation, and quantization to enable real-time edge inference.

Stage III (5–10 years): System integration and commercial validation.

Once accuracy and real-time requirements are satisfied, the primary challenge shifts to system-level integration, including closed-loop perception–control systems, hardware-level multi-sensor synchronization, and integration with farm management platforms. End-to-end frameworks proposed by Ni et al. and Cowie et al. provide early examples of such systems. The next step is large-scale deployment and economic validation in commercial farming environments.

5.3. Perspective: Towards Fully Autonomous Self-Supervised Agricultural Perception Systems

The following discussion presents our perspective on the long-term evolution of agricultural stereo vision toward fully autonomous, self-supervised systems. While grounded in the trends and gaps identified in the preceding sections, the proposals and projections made here are forward-looking and reflect our own research outlook, rather than a consolidated summary of existing experimental evidence.

The ultimate goal of agricultural stereo vision is to develop fully autonomous, self-supervised perception systems capable of continuous learning, adaptation, and performance optimization without human intervention.

Self-supervised learning is central to this vision. By exploiting stereo left–right consistency, temporal continuity, and multi-view geometric constraints, supervision can be derived without manual annotations. This enables continuous learning from large-scale field data across growth stages, weather conditions, and crop types. Methods based on 3DGS-generated supervision and 2DGS-driven data synthesis have already demonstrated the feasibility of this direction.

Vision foundation models represent another key pathway. The emergence of models such as Foundation Stereo suggests that large-scale pretraining can provide strong zero-shot generalization capability. Future agricultural perception systems may no longer require retraining for each new scenario but instead adapt rapidly to new tasks and environments. Synthetic environments such as UE5-Forest highlight cases where real-world depth acquisition is difficult due to thin, overlapping structures that challenge LiDAR and structured light systems—precisely where synthetic data and foundation models offer maximum value.

Fully autonomous systems should further incorporate continual learning, automatic failure detection, adaptive recalibration, and cross-domain knowledge transfer. This requires edge–cloud collaborative architectures, lifelong learning frameworks, and intuitive interfaces for non-expert users.

Despite challenges such as extreme environmental diversity, long-tail distributions, and engineering reliability, recent progress—from hundreds to tens of thousands of stereo pairs, and from pixel-level errors to sub-pixel accuracy—demonstrates the feasibility of this trajectory. Ultimately, the future of agricultural stereo vision will not be determined by breakthroughs in individual algorithms, but by the co-evolution of data, algorithms, hardware, and system integration. When these components converge into a unified intelligent system capable of perception, learning, and decision-making, stereo vision will evolve from a laboratory technique into a core productive force in modern agriculture.

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